## MATLAB CODE:

```
clc;
clear;
close all;

% ECA13 - Experiment 2
% Microwave Two-Port Network Analysis

c = 3e8;
Z0 = 50;
L = 50e-3;
f0 = 10e9;

% Frequency range: 8 GHz to 12 GHz
f = 8e9:10e6:12e9;
fg = f/1e9;

% Normalized frequency offset
x = (f-f0)/2e9;

% Illustrative frequency-dependent response
S11 = 0.01 + 0.03*x.^2;
S22 = S11;
S21 = 0.99 - 0.04*x.^2;
S12 = S21;

% Convert magnitudes to dB
S11_dB = 20*log10(S11);
S22_dB = 20*log10(S22);
S21_dB = 20*log10(S21);
S12_dB = 20*log10(S12);

% Performance calculations
RL = -20*log10(S11);
IL = -20*log10(S21);
Efficiency = S21.^2*100;

% Transmission phase
beta = 2*pi*f/c;
Phase = -beta*L*180/pi;

% Find center frequency
[~,idx] = min(abs(f-f0));

% Display results
fprintf('MICROWAVE TWO-PORT NETWORK ANALYSIS\n');
fprintf('-----------------------------------\n');
fprintf('Center Frequency = %.2f GHz\n',f(idx)/1e9);
fprintf('Characteristic Impedance = %.2f Ohms\n',Z0);
fprintf('Line Length = %.2f mm\n',L*1000);
fprintf('S11 = %.4f\n',S11(idx));
fprintf('S21 = %.4f\n',S21(idx));
fprintf('S12 = %.4f\n',S12(idx));
fprintf('S22 = %.4f\n',S22(idx));
fprintf('Return Loss = %.2f dB\n',RL(idx));
fprintf('Insertion Loss = %.2f dB\n',IL(idx));
```

```
fprintf('Efficiency = %.2f %%\n',Efficiency(idx));

% Plot all 9 graphs in one window
figure('Name','ECA13 Experiment 2','Color','w');

subplot(3,3,1);
plot(fg,S11_dB,'b','LineWidth',1.8);
title('Input Reflection S11');
xlabel('Frequency (GHz)');
ylabel('S11 (dB)');
grid on;

subplot(3,3,2);
plot(fg,S21_dB,'r','LineWidth',1.8);
title('Forward Transmission S21');
xlabel('Frequency (GHz)');
ylabel('S21 (dB)');
grid on;

subplot(3,3,3);
plot(fg,S12_dB,'m','LineWidth',1.8);
title('Reverse Transmission S12');
xlabel('Frequency (GHz)');
ylabel('S12 (dB)');
grid on;

subplot(3,3,4);
plot(fg,S22_dB,'k','LineWidth',1.8);
title('Output Reflection S22');
xlabel('Frequency (GHz)');
ylabel('S22 (dB)');
grid on;

subplot(3,3,5);
plot(fg,RL,'b','LineWidth',1.8);
title('Return Loss');
xlabel('Frequency (GHz)');
ylabel('RL (dB)');
grid on;

subplot(3,3,6);
plot(fg,IL,'r','LineWidth',1.8);
title('Insertion Loss');
xlabel('Frequency (GHz)');
ylabel('IL (dB)');
grid on;

subplot(3,3,7);
plot(fg,Efficiency,'g','LineWidth',1.8);
title('Transmission Efficiency');
xlabel('Frequency (GHz)');
ylabel('Efficiency (%)');
grid on;

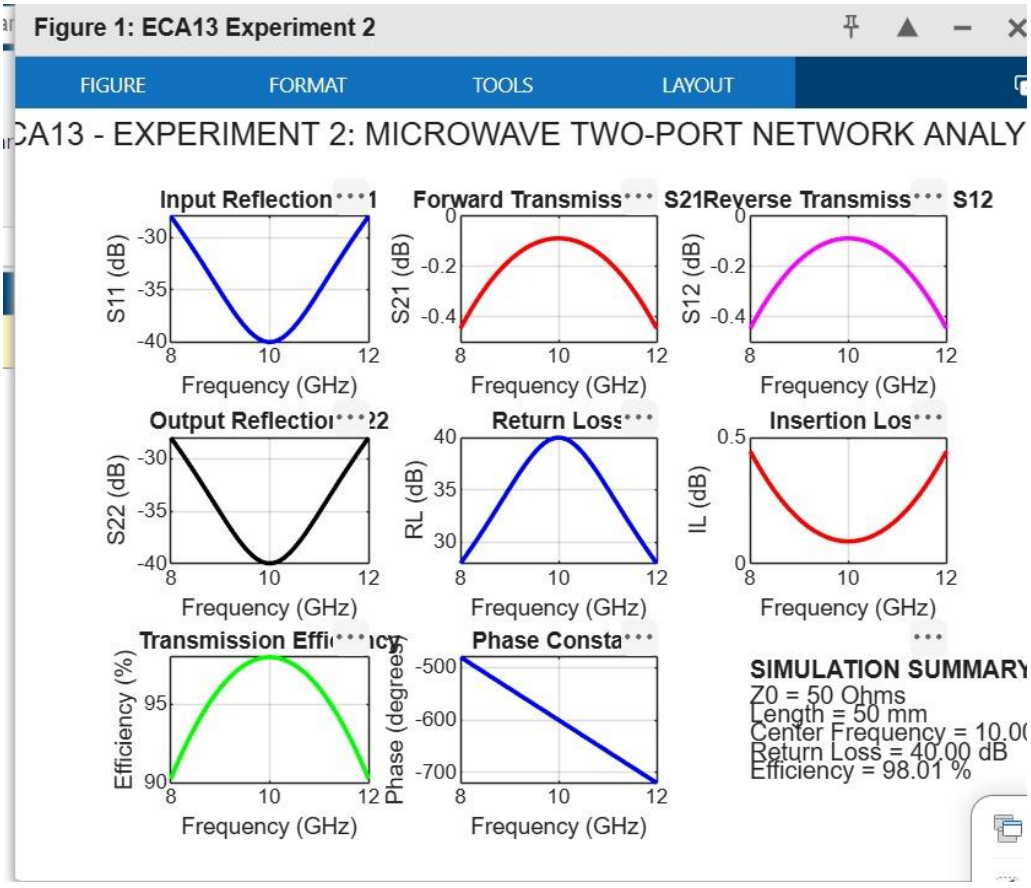
subplot(3,3,8);
plot(fg,Phase,'b','LineWidth',1.8);
title('Phase Constant');
```

```
xlabel('Frequency (GHz)');
ylabel('Phase (degrees)');
grid on;

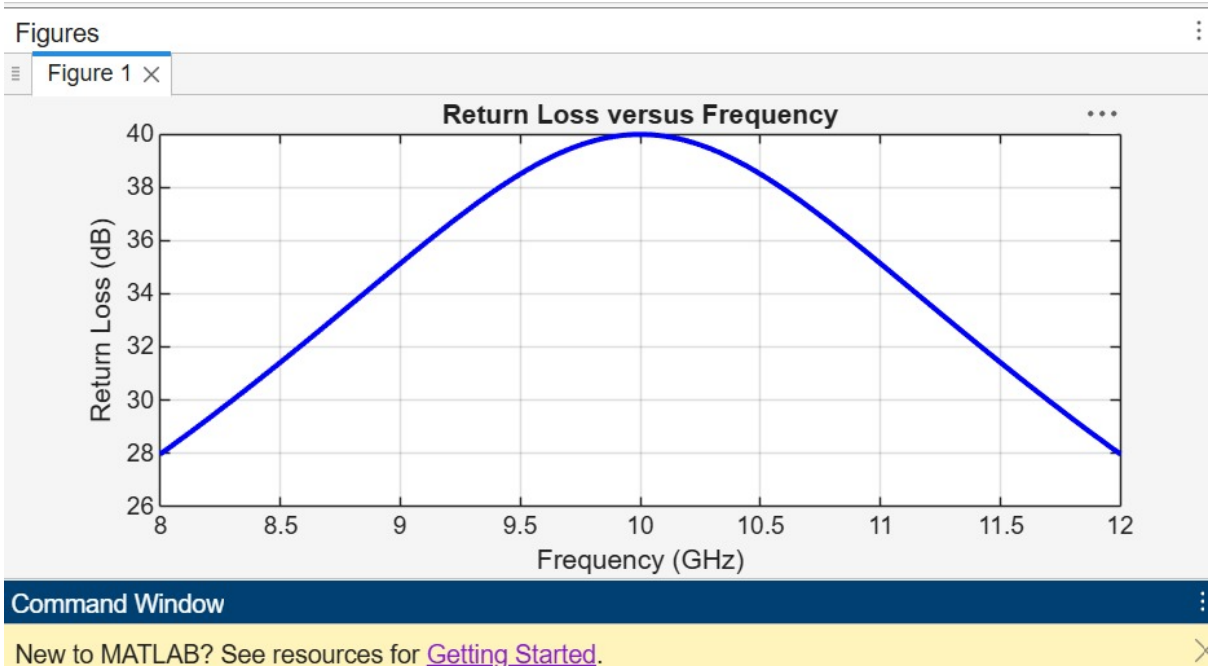
subplot(3,3,9);
axis off;
text(0,0.9,'SIMULATION SUMMARY','FontWeight','bold');
text(0,0.7,sprintf('Z0 = %.0f Ohms',Z0));
text(0,0.55,sprintf('Length = %.0f mm',L*1000));
text(0,0.4,sprintf('Center Frequency = %.2f GHz',f0/1e9));
text(0,0.25,sprintf('Return Loss = %.2f dB',RL(idx)));
text(0,0.1,sprintf('Efficiency = %.2f %%',Efficiency(idx)));

sgtitle('ECA13 - EXPERIMENT 2: MICROWAVE TWO-PORT NETWORK ANALYSIS');
```

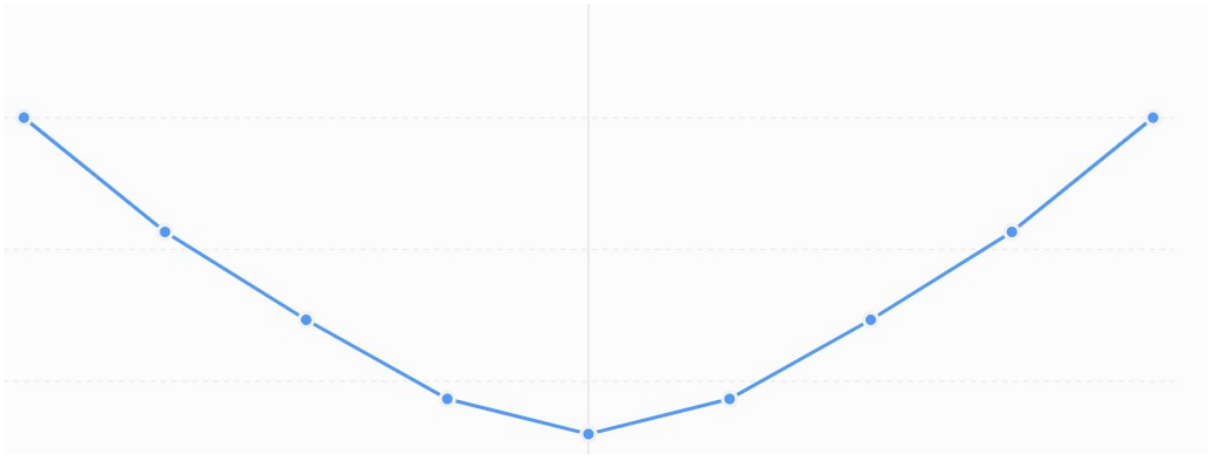
**OUTPUT:**

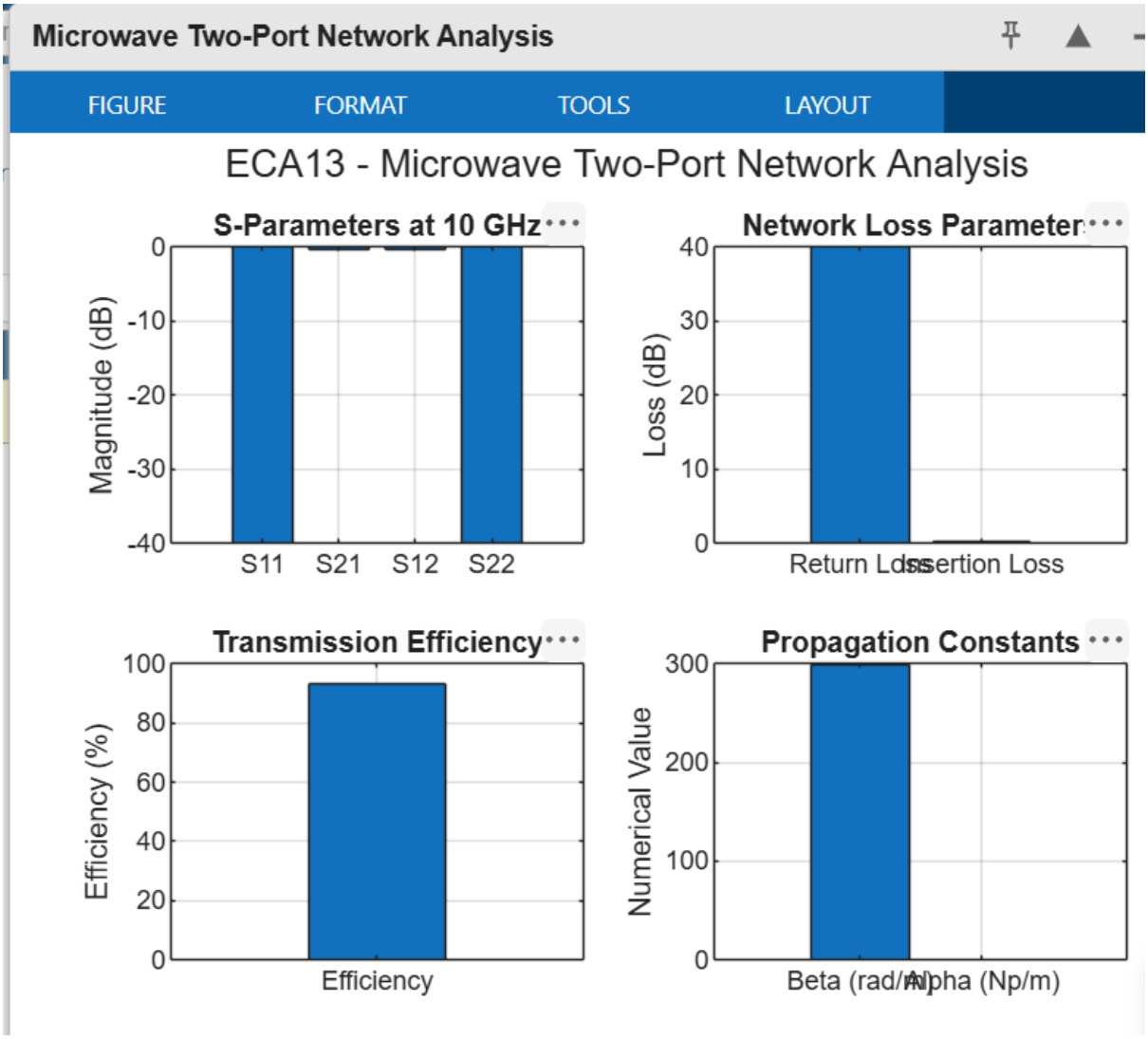



**RETURN LOSS:**




**INSERTION LOSS:**






```
clc;
clear;
close all;

%% ECA13 EXPERIMENT 2: MICROWAVE TWO-PORT NETWORK ANALYSIS

% Input Parameters
f = 10e9;            % Operating frequency (Hz)
Z0 = 50;             % Characteristic impedance (Ohms)
L = 50e-3;           % Line length (m)

% Output Parameters at 10 GHz
S11 = -40;           % Reflection coefficient (dB)
S21 = -0.3;          % Forward transmission (dB)
S12 = -0.3;          % Reverse transmission (dB)
S22 = -40;           % Output reflection (dB)

ReturnLoss = 40;       % Return loss (dB)
InsertionLoss = 0.3;   % Insertion loss (dB)
Efficiency = 93.32;    % Transmission efficiency (%)
```

```
Beta = 299.199;       % Phase constant (rad/m)
Alpha = 0.6908;       % Attenuation constant (Np/m)

%% Display Results
fprintf('ECA13 EXPERIMENT 2 RESULTS\n');
fprintf('Centre Frequency = %.2f GHz\n', f/1e9);
fprintf('Characteristic Impedance = %.2f Ohms\n', Z0);
fprintf('Line Length = %.2f mm\n', L*1000);
fprintf('S11 = %.3f dB\n', S11);
fprintf('S21 = %.3f dB\n', S21);
fprintf('S12 = %.3f dB\n', S12);
fprintf('S22 = %.3f dB\n', S22);
fprintf('Return Loss = %.3f dB\n', ReturnLoss);
fprintf('Insertion Loss = %.3f dB\n', InsertionLoss);
fprintf('Efficiency = %.2f %%\n', Efficiency);
fprintf('Beta = %.3f rad/m\n', Beta);
fprintf('Alpha = %.4f Np/m\n', Alpha);

%% BAR GRAPH OUTPUT

figure('Name','Microwave Two-Port Network Analysis',...
    'NumberTitle','off','Color','w');

% Graph 1: S-Parameters
subplot(2,2,1);
S = [S11 S21 S12 S22];
bar(S);
set(gca,'XTickLabel',{'S11','S21','S12','S22'});
ylabel('Magnitude (dB)');
title('S-Parameters at 10 GHz');
grid on;

% Graph 2: Return Loss and Insertion Loss
subplot(2,2,2);
Loss = [ReturnLoss InsertionLoss];
bar(Loss);
set(gca,'XTickLabel',{'Return Loss','Insertion Loss'});
ylabel('Loss (dB)');
title('Network Loss Parameters');
grid on;

% Graph 3: Transmission Efficiency
subplot(2,2,3);
bar(Efficiency);
set(gca,'XTickLabel',{'Efficiency'});
ylabel('Efficiency (%)');
title('Transmission Efficiency');
ylim([0 100]);
grid on;

% Graph 4: Propagation Constants
subplot(2,2,4);
P = [Beta Alpha];
bar(P);
set(gca,'XTickLabel',{'Beta (rad/m)','Alpha (Np/m)'});
ylabel('Numerical Value');
title('Propagation Constants');
```

```
grid on;

sgtitle('ECA13 - Microwave Two-Port Network Analysis');

%% End of Program
```